

MLFF-DATA9A9b Production DATA6-DATA8 Materialization

mdstats project

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1. Purpose

DATA9A9b closes the orchestration gap between a completed checkpoint-bound DATA6 model sweep and executable final/fold DATA8 training jobs. DATA7 and DATA8 scientific semantics already exist. This stage makes their production realization exact, restartable, tamper-evident, and bound to one replay corpus.

The stage owns neither feature definitions nor MACE training. It owns:

- freezing the complete DATA6 sweep identity;
- freezing every canonical final and cross-validation DATA7 domain;
- freezing all DATA7 fitting, objective, weighting, checkpoint, and selection policies;
- binding one exact replay-train/replay-monitor plan;
- checkpointing each realized DATA7 bundle independently;
- invalidating downstream DATA8 evidence when any DATA7 artifact changes;
- promoting a complete final/fold DATA8 tree only after all prerequisites pass;
- emitting one immutable materialization record that can update production qualification.

DATA9B remains closed until a complete production DATA9A9b record exists.

2. Ownership boundary

`mdstats.training_data.production_materialization` owns orchestration and restart evidence only.

- DATA6 owns checkpoint-bound descriptors, predictions, residuals, and blinded summaries.
- DATA7 owns fold-local feature fitting, E0 fitting, objectives, weights, selection, and coverage.
- DATA8 owns extxyz export, MACE configuration, replay staging, loader dry runs, and sealed evaluation metadata.
- Replay owns replay-file inspection, disjointness, label provenance, and retention policy.
- DATA9B owns training execution, checkpoint comparison, committee construction, and freeze.

The production orchestrator calls these owners and records their native bundle digests. It does not reproduce their algorithms.

3. Frozen production plan

`ProductionMaterializationPlan` binds:

- source, frame, DATA4, DATA5, and DATA6 identities;
- the complete DATA6 model-sweep checkpoint;
- exact descriptor and prediction manifests;
- every canonical `FeatureFitDomain` from DATA5;
- feature-metric, atomic-reference, objective, configuration-weight, checkpoint-metric, and selection-budget policies;
- the exact foundation checkpoint;
- optional foundation atomic reference energies;
- MACE compatibility policy and passing source probe;
- the exact replay train and monitor artifacts;
- optimizer, checkpoint-control, and extxyz policies;
- selected DATA7 ladder size and replay exposure settings.

The canonical domain set contains exactly one final-development domain and one training domain for each declared cross-validation fold. The plan is rejected when the replay plan is unresolved, when the completed DATA6 sweep differs from the DATA6 bundle, or when the DATA8 foundation checkpoint differs from the checkpoint used for DATA6 predictions.

The current plan schema is `mdstats.production-materialization-plan.v10`. Current TRAIN2 target-size-prefix authority is independent of foundation-head topology: a genuinely single-head foundation carries no selected-head qualification, while an inspected multi-head foundation still requires the qualified EXTRACT1 selected-head training realization. Legacy v9 plans remain readable and retain their historical requirement for selected-head qualification.

4. Restart semantics

Each DATA7 domain is written independently to `data7/<domain_digest>.data7.zip`. `ProductionData7ArtifactRecord` binds:

- domain digest;
- DATA7 bundle digest;
- safe relative path;
- file SHA-256.

`ProductionMaterializationCheckpoint` is atomically replaced after every newly committed canonical domain. A bounded run may stop after `max_new_data7_domains`; shared-cache hits count toward that limit because the limit governs checkpoint advancement, not expensive-computation count. A later run verifies and reuses all valid records. Parallel domain completion may be out of order, but checkpoint mutation remains in canonical plan order.

The production implementation SHALL avoid hidden repeated whole-corpus work:

- build the frame-array index once and reuse it across DATA7 domains;
- use authenticated columnar/shard-batched raw descriptor sources where available while keeping every domain's fitted scaler/PCA/E0/weight state independent;
- admit independent final/fold domain fits through the runtime resource queue using the campaign CPU budget, a conservative peak incremental-memory estimate, inner native widths of one, and no GPU jobs;
- keep mutable extraction caches task-local under concurrent fitting;
- have workers publish immutable authenticated DATA7 cache generations and return compact receipts while the coordinator alone mutates production records/checkpoints;
- reuse lineage-identical DATA7 artifacts across seeds and modes through the shared recipe cache; current cache writes use atomic content-addressed generations while legacy flat generations remain read-compatible;
- verify and return each promoted DATA7 bundle in one load path rather than hashing/parsing it twice;
- derive canonical final/CV feature-fit domains inside the materialization-plan builder and reject duplicate topology keys, missing final-development domains, or CV topology that differs from the authenticated CV plans;
- resolve target-size prefixes lazily by canonical training-domain digest while confining synthetic coverage-authority IDs to REPAIR2 lookup; derive each pre-selection evaluation cohort from the authenticated final coverage domain minus the maximum-qualified prefix;
- authenticate shared fitted-core results separately from their execution recipe, publish fitted-core indices create-once/validate-winner, and reject divergent full-DATA7 or fitted-core results for one recipe;
- treat stale reconstructible fitted-core indices as cache misses and fall back to a fresh fit after exact lineage/foundation-input validation;
- admit authenticated fitted-core reuse with a conservative reuse-path memory estimate rather than the full feature-fit peak.

These rules make post-DATA6 orchestration linear in frame count for fixed fold count, feature dimension, and largest selection ladder, apart from bounded $O(K^2d)$ selected-neighbor reporting and fixed-dimension linear algebra.

If a DATA7 artifact is missing, malformed, modified, or bound to a foreign DATA6 bundle, it is either rejected or recomputed according to the explicit execution policy. Any invalid DATA7 artifact invalidates the DATA8 tree because the training protocol depends on its fitted transforms, E0 values, selection, and weights.

5. Exact replay binding

Production multihead replay requires two disjoint files:

- replay training data;
- replay monitor data.

The plan binds their configuration counts, atomic-number sets, geometry identities, numerical energy/force/stress label identities, label keys, label provenance, foundation-checkpoint identity for pseudo-labels, selection provenance, head weight, target weight, and retention policy.

DATA8 may stage and rescale the replay-train file to realize fixed-file MACE weights. Therefore source and staged file byte hashes need not be equal. The orchestrator verifies semantic equivalence: configuration identity, labels, provenance, counts, species, and replay policy must remain unchanged.

6. DATA8 promotion

DATA8 construction begins only when every planned DATA7 domain is valid. Immutable fixed-file recipes are enumerated and deduplicated before production-tree mutation. Cache misses may be populated by balanced fresh CPU-only interpreter batches using compact recipe/path descriptors and mmap/file-backed read-only context when the estimated output volume is sufficient to amortize fresh-interpreter startup; smaller batches use the serial producer. CPU, RAM, task count, the measured/estimated worker-context spill, and the configured free-disk reserve bound this execution-only concurrency. If the transient context cannot coexist with the required immutable outputs and reserve, DATA8 falls back to the serial producer before launching subprocesses. Workers publish only authenticated fixed-file cache generations; the parent then assembles the production tree canonically in hidden staging, verifies it, moves it into a content-addressed generation directory, and exposes it by an atomic data8 symlink switch. The output tree contains:

- one final-development job;
- one job per cross-validation fold;
- selected target-train and target-monitor extxyz files;
- fold evaluation files that are never referenced by training configs;
- staged foundation and replay resources;
- MACE configs, commands, job manifests, and loader dry-run evidence;
- sealed locked-test metadata without materialized locked-test labels.

A DATA8 artifact record binds the native DATA8 bundle digest and a deterministic relative-path/file-SHA tree manifest. Fixed-file cache population is reconstructible execution state and never replaces this promoted authority. Externally owned foundation, selected-head, and replay sources are authenticated before crossing an inode-independent copy boundary into mdstats-owned content-addressed snapshots. Variant trees may hardlink-or-copy only from those owned immutable snapshots/caches, so later external in-place mutation cannot change an existing materialization. Repeated weighted-replay byte realizations and MLCV TRUE_DFT replay-light monitor realizations may be reused from exact recipe-bound execution caches. Completion is promoted only after the whole staged tree is hashed and its bundle is valid. Restored records perform independent tree verification. Failed construction removes an unpromoted partial DATA8 directory and records failure evidence.

7. Atomic-reference modes

For foundation-model fine-tuning, production normally uses FOUNDATION_RESIDUAL. In that mode every fit-domain frame requires a persisted foundation energy prediction, the exact foundation checkpoint digest, and a complete foundation E0 mapping for all target elements.

From-scratch total-energy E0 fitting remains available for explicit tests and non-transfer workflows. Foundation predictions or E0s must not be supplied to a from-scratch fit.

8. Materialization record

ProductionMaterializationRecord exposes:

- plan and checkpoint identity;
- ordered DATA7 bundle digests;
- DATA8 bundle digest;
- completion state;
- verified loading of native DATA7 and DATA8 bundles.

The filesystem root is a relocatable location hint and is excluded from the scientific content digest. The record supplies the exact DATA7/DATA8 digests and replay binding needed by `ProductionCorpusQualificationRecord`. The production qualification builder accepts the complete record directly, verifies DATA6 lineage, loads the native bundles, and derives replay/materialization status without caller-supplied duplicate digests.

9. Failure rules

The stage fails closed for:

- incomplete or foreign DATA6 model-sweep evidence;
- changed source, frame, DATA4, DATA5, or DATA6 identity;
- a foundation checkpoint mismatch;
- unresolved or absent replay when replay is required;
- foreign or noncanonical DATA7 domains;
- modified DATA7 files;
- modified DATA8 files or tree membership;
- missing MACE compatibility support;
- selection sizes absent from a DATA7 ladder;
- replay/target exact geometry overlap;
- incomplete residual-E0 evidence;
- any attempt to materialize DATA8 before all DATA7 domains pass.

10. Focused acceptance tests

The release gate must cover:

1. bounded one-domain execution followed by deterministic resume;
2. one final and all fold-local DATA7 bundles;
3. exact replay binding in every DATA8 protocol;
4. no evaluation or locked-test file in a MACE training configuration;
5. DATA7 tamper detection and downstream DATA8 invalidation;
6. DATA8 tree tamper detection;
7. plan mismatch rejection;
8. historical DATA6 compatibility remaining unchanged;
9. source/wheel export and registry parity;
10. immutable record serialization and relocation-safe identity;
11. exact bounded-FPS prefix equivalence to the former deterministic algorithm;
12. descriptor-summary reuse across overlapping DATA7 domains;
13. shared DATA7 reuse across training variants and process restarts;
14. no duplicate parse/hash pass when verified DATA7 immediately feeds DATA8.
15. target-size candidate-prefix planning scales with unique domain/size pairs rather than development-frame count times optimizer variants;
16. target-size size changes reuse an authenticated selection-invariant DATA7 fitted core while producing the exact same size-specific DATA7 bundle digest as a clean refit;
17. current DATA7 shared-cache recipe identity excludes DATA8-only evaluation membership and target-study outcome state, with legacy v1 recipe reads preserved;
18. the real campaign preparation bridge reaches DATA7 for folds=0 screening, canonical selected CV, and per-seed selected CV without synthetic-ID namespace leakage;
19. malformed plan topology (missing final domain, duplicate topology key, or wrong CV fold membership) fails before DATA7 execution;
20. same-recipe divergent full DATA7 and fitted-core results fail closed, including concurrent fitted-core publishers;
21. stale fitted-core carriers fall back to fresh fitting and reuse-path RAM admission is evaluated independently from full-fit RAM.

11. Scope of the current implementation

Version 0.20.55a0 retains and hardens the complete restartable control layer and qualifies it on a bounded four-domain target/replay workflow. It does not claim that the full 2,734-frame production sweep has completed in this execution environment. The production corpus must run DATA9A9a to completion, then execute this DATA9A9b plan before DATA9B training begins.